

Status: Submitted ▾

Mentor:  Person

GROUP PROJECT PROPOSAL

Project Title: Visual Content-Based Music Recommendation

1. Team Information

Name	StudentId/Email	Role
THU MAI HUYNH MINH	104240759	Main Researcher/AI models
An Nguyen Hoang Gia	104240384	G1 (System + UI)
LINH TRAN GIA	104240762	G2 (Data + Evaluation)
Tran Nguyen Bao	104240291	G1 (System + UI)
Vy Nguyen Ngoc Minh	104240475	G2 (Data + Evaluation)
Tram Ha Ngoc Bao	104240468	G2 (Data + Evaluation)

2. Project Idea Summary

In 2–3 sentences, describe your project idea.

Recently there is a feature on social media apps, beginning with Tiktok and followed by Instagram, that automatically selects an appropriate song when editing an image that the user uploads based on its aesthetic, mood, themes and visuals. Oftentimes, the selected song's emotional tone and thematic tags are suitable with the content of the image. In this project, we aim to resemble that feature in a much simpler way with a small-scale AI system.

3. Problem Statement

What problem are you trying to solve?

Our project aims to reduce the pain of manually selecting songs when uploading images, making the process faster and much more efficient.

Why is this problem important or useful (optional)?

This is useful because it helps users to save time to find music that matches their content of images or videos and possibly increase the user's overall satisfaction with the website. Based on AI recommendations, users also can improve creativity in their content creation.

4. AI / Data Science Component

What AI / Data Science method(s) might be used?

- ☐ Classification
- ☐ Regression
- ☐ Clustering
- ☒ Recommendation System
- ☒ Computer Vision
- ☒ Natural Language Processing
- ☐ Time Series Analysis
- ☐ Generative AI
- ☐ Other: ____

Briefly explain how AI / Data Science is used in your project.

The process should contain the following steps:

1. Understand the image (Computer Vision)
2. Interpret its mood (Machine Learning, NLP)
3. Matching it to songs (Recommendation Systems)

First, computer vision is applied to extract meaningful features from images (such as images, texts, color themes...) and understand their visual content. Then we use supervised learning or NLP (for texts) to predict the mood of the image (feature vector for the model and explicit label for the web app, because humans understand words 😊). Finally we use a recommendation system to match songs that have similar features to the image.

5. Data

What kind of data will you use?

Image data:

- Public dataset

- Our own collected images
- Small curated dataset

Song data:

- Manual labeling
- Spotify API

Text data:

- Lyrics/lyrics description (Musixmatch, Genius)

Feature data (derived data):

- Features from songs and images

Evaluation data

Possible data source(s):

- ☒ Public dataset
- ☒ Collected manually
- ☒ Web scraping / API
- ☒ Synthetic / generated data
- ☐ Other: _____

Example dataset(s) or source(s) (if known):

Lyrics-based Datasets

- Musixmatch dataset
 - + Large scale datasets (often paired with Spotify IDs)
 - + Limitation: often contains bag-of-words only, not full lyrics (copyright)
- Genius (via API or scraping)
- Million Song Dataset (MSD)
 - + Audio + metadata

Audio features datasets

- Spotify API
- DEAM dataset (Database for Emotion Analysis in Music)

Tag/metadata datasets

- [Last.fm](https://last.fm/) datasets

Manual mood tags

...

6. Expected Output

What will your project produce?

- ☐ Prediction model
- ☐ Dashboard / visualization
- ☒ Web / mobile app prototype
- ☐ Recommendation tool
- ☐ Chatbot / assistant
- ☒ Report (MUST HAVE)
- ☒ Code (MUST HAVE)
- ☐ Other: _____

Describe the final output in 1–2 sentences.

The final output should present a web application that mimics social media apps UI (e.g.: Tiktok, Instagram,...) where the user uploads an image and that image will instantly be matched with a song (or list of suitable songs) that matches with the image's mood and aesthetics. The web will also display detected mood/style.

The flow should look like this:

1. user opens the website
2. uploads an image
3. the system analyzes the image
4. the app shows the detected mood/style
5. the app recommends a few matching songs

7. Target Users / Audience (optional)

Who will benefit from this project?

Content creators, app users.

How will they benefit?

They can quickly find mood-matched background music for photos, videos, or social media posts.

8. Feasibility

Briefly describe your team implementation plan

Phase 1: Data collection and manual labeling

Phase 2: Image preprocessing and visual extracting using CLIP or ResNet

Phase 3: Core AI system building

Phase 4: Web app development

Phase 5: Testing and evaluation

Component	Method	Metric
Vision	tag/embedding quality	accuracy/neighbor consistency
Mood	predictions vs. human labels	F1/MAE
Recommendation	human rating	avg score
Recommendation	ranking	top-k accuracy

Phase 6: Refinement

Literature review

Main tools / technologies you may use:

- Python (main programming language)
- PyTorch or TensorFlow (for deep learning models)
- OpenCV / PIL (image processing)
- CLIP / pretrained CNN (ResNet, EfficientNet) (image feature extraction)
- Scikit-learn (classification, similarity, evaluation)
- Pandas / NumPy (data processing and feature handling)
- CSV / JSON (storing song metadata)
- Flask or FastAPI (backend API)
- React (for a more advanced UI)
- External APIs
- Spotify API (song metadata, audio features)

9. Possible Challenges

List 2–3 challenges you may face.

1. Limited dataset.
 2. The same emotion can be understood differently by each person, matching is mostly subjective.
 3. Manual labeling can be inconsistent
 4. Difficulty in connecting AI with the web app
 5. Evaluation difficulty, the model may produce confusing, abstract or multiple answers at the same time
 6. Hard to match visual features with musical features
-

10. Success Criteria

How will you know the project is successful?

- ☒ Model has acceptable performance
- ☐ Useful insights are found from data
- ☒ Prototype works correctly
- ☒ Results can be clearly explained
- ☐ Other: _____

One sentence summary of success:

The project is successful if the system is able to recommend music that somewhat appropriately matches the mood of input images without severe errors, smooth user experience, clear presentation of results, and a working integration between the AI model and the web interface.